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Layer-to-Layer Thermal Runaway Propagation of Open-Circuit Cylindrical Li-ion Batteries: Effect of Ambient Pressure

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Abstract

Preventing thermal runaway propagation is crucial to ensure the safety of lithium-ion battery energy storage system, especially in low-pressure air-transport and the near-vacuum space environment. This work investigates the linear thermal-runaway propagation in $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$ 18650 cylindrical battery layers under ambient pressure from 0 atm to 1 atm. Results indicate that the 1-D layer-to-layer thermal runaway propagation rate decreases with decreasing SOC and ambient pressure. As the SOC decreases from 100% to 30%, the thermal runaway propagation rate decreases from 1.73 [layer/min] to 0.30 [layer/min] at 1 atm. For 30% SOC cells, the thermal runaway propagation rate decreases by about 23% as the ambient pressure decreases from 1 atm to 0.2 atm and eventually drops to zero at 0 atm. The X-ray computed tomography imaging reveals that low pressure can weaken both external flaming combustion and internal thermal runaway reactions during the venting stage. As the ambient pressure decreases, such dual effect increases the thermal runaway temperature from 200 to 310 °C, reduces the maximum surface temperature from 800°C to 400 °C, and lowers the burning mass loss fraction from 32% to 10%. Finally, a simplified heat transfer model is proposed to explain the effects of SOC and ambient pressure on thermal runaway propagation. These findings provide a new way to mitigate the thermal runaway propagation and help to assess the safety of battery piles in storage and transport.

Keywords: *Lithium-ion battery; low pressure; thermal safety; propagation rate; vacuum.*

1. Introduction

Lithium-ion battery (LIB) is one of the most promising energy storage technologies due to its high energy density and environmental friendliness [1,2]. Under the international push toward carbon neutrality targets, the demand for LIBs has risen rapidly. The global LIB market size was around USD 53.6 billion in 2020, estimated to grow at a compound annual growth rate of 19.0% from 2020 to 2028 [3]. In pursuit of large capacity and high energy density, more and more active materials are developed and applied in LIBs [4]. These reactive components reduce the thermal stability of LIBs and inevitably induce potential thermal failures. Especially when LIBs are subject to external abuse conditions or internal flaws, thermal runaway will be easily triggered [5,6]. Thermal runaway of LIBs can release a large amount of heat and eject flammable materials, which may further lead to flaming combustion [7]. Numerous LIB fire accidents have been reported in recent years, causing enormous economic losses and posing a severe challenge to public safety [8].

In storage and transport, massive LIBs are tightly stacked [9]. Once a single LIB cell undergoes thermal runaway, the released heat could easily propagate to the adjacent ones, inducing the thermal runaway propagation in the entire battery pile. Such a cascading failure can exponentially increase the fire hazards and even cause an explosion [10]. Although statistically rare, LIB fires exhibit significantly different from other fire hazards in the initiation route, spread rate, duration, toxicity, and suppression [11]. Thus, considerable research efforts have been devoted to investigating the characteristics of thermal runaway propagation and developing mitigation strategies accordingly. In literature, the effects of LIB chemistry [12–15], state of charge (SOC) [16,17], LIB form factors [18,19], connection modes [20,21], and cell arrangements [22,23] on thermal runaway propagation have been comprehensively investigated. To alleviate the thermal runaway propagation, most studies mainly focused on the active extinguishing methods [24] or passive mitigating strategies [25,26]. For example, agents like CO_2, HFC-227ea [27], Novec 1230 [28], $C_6F_{12}O$ [29], and water mist [30] in suppressing LIB fire has been extensively studied. Meanwhile, several works employed thermal insulation materials [31,32] or optimized phase change materials [33] in LIB modules to inhibit thermal runaway propagation passively.

The aforementioned passive approaches show remarkable mitigation efficacy, especially for the operating-state LIB modules. However, most of these strategies employ additional devices, which could occupy a lot of space and reduce the transport efficiency for LIBs. More effective mitigation methods for thermal runaway propagation are still needed for the safe storage and transport of LIBs. Considering that flaming combustion can accelerate thermal runaway propagation in cylindrical LIB modules [34], Weng et al. [35] decrease the ambient oxygen level and successfully mitigate the battery fire propagation. Inspired by their research, reducing the ambient pressure would be an alternative way to alleviate LIB fire. The reason could be that the flaming combustion after battery thermal runaway would be weakened under low ambient pressure [36–40]. Specifically, Fu et al. [36] found that the peak flame temperature of a single battery fire decreased from 590 °C to 300 °C as the pressure reduced from 101 kPa (1 atm) to 40 kPa (0.4 atm). Similarly, Xie et al. [38] calculated that the heat release rate of a singly LIB fire decreased from 1.12 kW to 0.67 kW, with the pressure decreasing from 0.95 atm to 0.2 atm. The above studies mainly focused on the combustion behavior of the single cell, while very few studies investigated the thermal runaway propagation under low pressure. Recently, Jia et al. [41] found that the thermal runaway propagation of 100%-SOC ternary batteries at 0.35 atm would be much slower. However, there is no research studying the thermal runaway propagation of low-SOC batteries under low pressure, whereas low-SOC is required for batteries in air transport. And the literature still lacks the experimental data and investigation of thermal runaway characteristics under the vacuum condition (0 atm).

To bridge the knowledge gap, a previously developed low-pressure chamber was employed to study the effectiveness of the depressurized method in mitigating battery fire propagation. By controlling the air inlet and outlet, the ambient pressure ranged from 0 atm to 1 atm. The LIB module was constructed from six cylindrical cells where the thermal runaway was initiated by a 100-W heater. The characteristic temperature of LIBs and ambient pressure were measured and systematically discussed. X-ray computer tomography imaging was used to indicate the internal structure changes of LIBs after the thermal runaway. Finally, a simplified heat transfer model was proposed to qualitatively elucidate the pressure effect on 1-D layer-to-layer thermal runaway propagation.

2. Experimental Methods

2.1 LIB samples

Typical 18650-type cylindrical batteries made by Samsung SDI Co., Ltd (ICR18650–22F) were tested. The cathode material of this LIB is $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$ (NCM523), where Nickel (Ni) is related to its energy density, Cobalt (Co) is responsible for the structural stability and cycling performance, and Manganese (Mn) is used for safety promotion. Intercalation graphite is the anode material of this LIB sample. The LIB cell has a nominal capacity of 2.2 Ah, a nominal voltage of 3.6 V, and an initial mass of 41.05 ± 0.05 g. Before testing, the SOC levels of all LIB cells were calibrated to 100% SOC and 30% SOC using the Neware battery test system (5 V/12 A, ± 6 mA). According to manufacturer recommendations, the key parameters for SOC calibration are listed in Table 1. To minimize the experimental uncertainty caused by LIB non-uniformity, the cells with the same voltage were selected for experiments after the charging and discharging process.

Table 1. Charge-discharge parameters of the LIB cells during SOC calibration.

Nominal capacity (Ah)	Nominal voltage (V)	Charging cut-off voltage (V)	Discharging cut-off voltage (V)	Charging & discharging current (A)	Cut-off current in constant-voltage mode (A)
2.2	3.6	4.20 ± 0.05	2.75 ± 0.05	1.10 ± 0.05	0.110 ± 0.005

2.2 Experimental setup

Six LIB cells were manually arranged as a rectangle battery pile, where a cylindrical heater was placed on the left side to trigger the thermal runaway. Note that the actual battery stockpile has numerous cells so that thermal runaway would have a 3D propagation path with complex characteristics. For the first time to investigate thermal runaway propagation under the vacuum condition, a simplified battery pile configuration was employed to highlight the pressure effect. The heater and sample cell have the same dimension (18 mm in diameter and 65 mm in height). From left to right, the battery pile has three layers. Thin stainless-steel (SS) wires (1.2 mm in diameter) were used to fix the entire battery pile and ensure good contact between the electric heater and the Layer-I cells (Fig. 1a). To acquire the accurate temperature and forestall the external short circuits, the body packaging of each cell was stripped off except for the plastic packaging of the cathode cap (around 15-mm height). Six K-type thermocouples were attached to the half-height of cell surfaces to measure the battery temperature (T_1-T_6). Each thermocouple had a tiny bead of 0.2 mm in diameter and an accuracy of 0.1 °C. The temperature information was collected by a digital data logger (HIOKI LR8400) with a sampling frequency of 1 Hz. A DC power supply (MS-DMP-D) was used to control the heating power of the cylindrical heater.

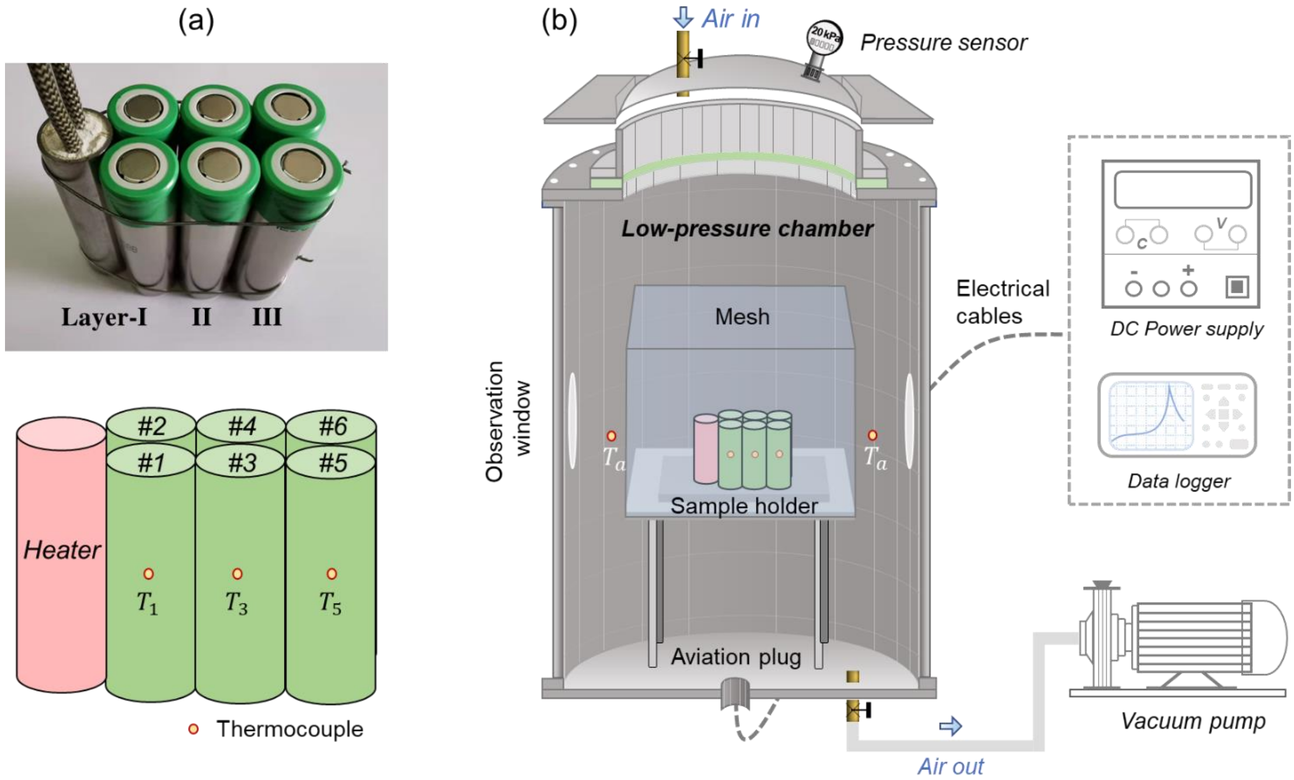

Fig. 1. (a) The tested LIB pile and the thermocouple arrangement, as well as the (b) experimental setup.

All tests were implemented in a previously designed low-pressure chamber built by SS panels with heat-insulating paint on the surface. The chamber had an internal volume of 260 L and could provide a well-controlled environment for thermal runaway propagation tests. As shown in Fig. 1b, a vacuum pump with an extraction rate of 600 L/min was attached to the low-pressure chamber. By controlling the air outlet and air inlet, this chamber can maintain a constant ambient pressure ranging from 0 atm to 1 atm. It should be noted that the low-pressure chamber is not a dynamic chamber with continuous air in and out. The sealed chamber designed in this work is enclosed so that no airflow could affect the thermal runaway propagation during the tests. Inside the chamber, a cuboidal SS mesh enveloped the LIB pile, preventing the observation windows from potential damage caused by the battery fire. The internal pressure was measured by a pressure sensor (Meacon MIK-P300G), and the internal ambient temperature (T_a) was monitored by two K-type thermocouples in the chamber. To ensure the airtightness of the low-pressure chamber, electric cables (i.e., power supply wires and thermocouples) were welded to the aviation plug and then passed through the chamber bottom (Fig. 1b). A digital camera (SONY FDR-AX40, 25 fps) was used to monitor the experimental phenomena. Before and after the experiment, the cell mass was measured by an electronic balance with an accuracy of 0.1 g.

2.3 Testing procedures and controlling parameters

The experiments were conducted under the ambient temperature of 30 ± 2 °C. Before the tests, the fixed LIB pile was carefully placed on the sample holder. After checking the working state of thermocouples, the low-pressure chamber was sealed. Then, the vacuum pump was switched on until the ambient pressure dropped to the desired value. The ambient pressure would be stable for at least 10 min before the heating test started. As demonstrated in Fig. 1, a cylinder heater with a constant heating power of 100 W was used to trigger the thermal runaway of Layer-I cells. Once the thermal runaway of LIB1 or LIB2 was initiated, the heater was turned off. During the tests, two key parameters were controlled:

(I) Ambient pressure. The experiments were conducted under the four pressures of 0 atm, 0.2 atm, 0.6 atm, and 100 atm. In the preliminary tests, the gases generated by the thermal runaway of the fully charged cell would lead to a maximum pressure increase of 0.007 atm. Afterward, the pressure could gradually decrease. Thus, the chamber volume and the selected pressure interval are large enough to ignore temporary and slight pressure rise caused by thermal runaway.

(II) SOC level. Two SOC values of 30% and 100% were selected in this work. Generally, LIB with a higher SOC value has a higher fire risk, so the SOC level for LIBs in air transport is required to be less than 30% [42]. Therefore, cells with 30% SOC were tested to study the thermal runaway propagation of cells in storage and transport. Meanwhile, the fully charged LIB (100% SOC) was considered the extreme case and tested in this work.

The tests were stopped once the LIB temperature dropped to its initial value. To reduce the experimental uncertainty, all thermal runaway propagation tests were repeated at least two times. After the experiments, X-ray computed tomography (CT) was employed to study the structural characteristics of the burnt cells.

3. Results and Discussions

3.1 Thermal runaway phenomena at normal pressure

Fig. 2a shows the typical thermal runaway phenomena of fully charged (100% SOC) cells at 1 atm. Initially, the LIB module was heated by a cylindrical heater. During the heating, the positive cap of LIB-1 slightly swelled, indicating the occurrence of side reactions accompanied by gas generation inside the battery. As internal side reactions proceeded, more and more gases were generated, resulting in the pressure increase inside the batteries. At around 6 min, the safety valve of LIB-1 broke, along with the ejection of little gas and insulation materials. This is because the internal pressure of LIB-1 surpassed the threshold of the safety valve (1.5 MPa – 2 MPa) [43,44]. After a few seconds, the safety venting of LIB-2 happened. Without the protection of safety valve, gas release can be clearly observed. At about 7 min, the thermal runaway of LIB-1 was triggered, marked by the ejection of a large amount of white smoke and sparks. These sparks may consist of aluminum and lithium carbonate (Li_2CO_3) particles, which might be produced by the decomposition of battery components [45]. Finally, emitted gases were ignited by hot sparks, forming a violent jet flame. The interval time between LIB-1 thermal runaway and LIB-2 thermal runaway was less than 10 s.

For battery pile with 30% SOC, the thermal runaway process of Layer-I cells at 1 atm is demonstrated in Fig. 2b. Similarly, the safety venting of LIB-1 and LIB-2 successively occurred with the loud sounds. However, no spark ejection and flaming combustion can be observed, indicating the weak thermal runaway intensity of low-SOC cells. Since the vast smoke reduced the visibility inside the low-pressure chamber, the subsequent thermal runaway propagation cannot be directly observed. Thus, the temperature information was employed to analyze the overall characteristics, which will be comprehensively discussed later.

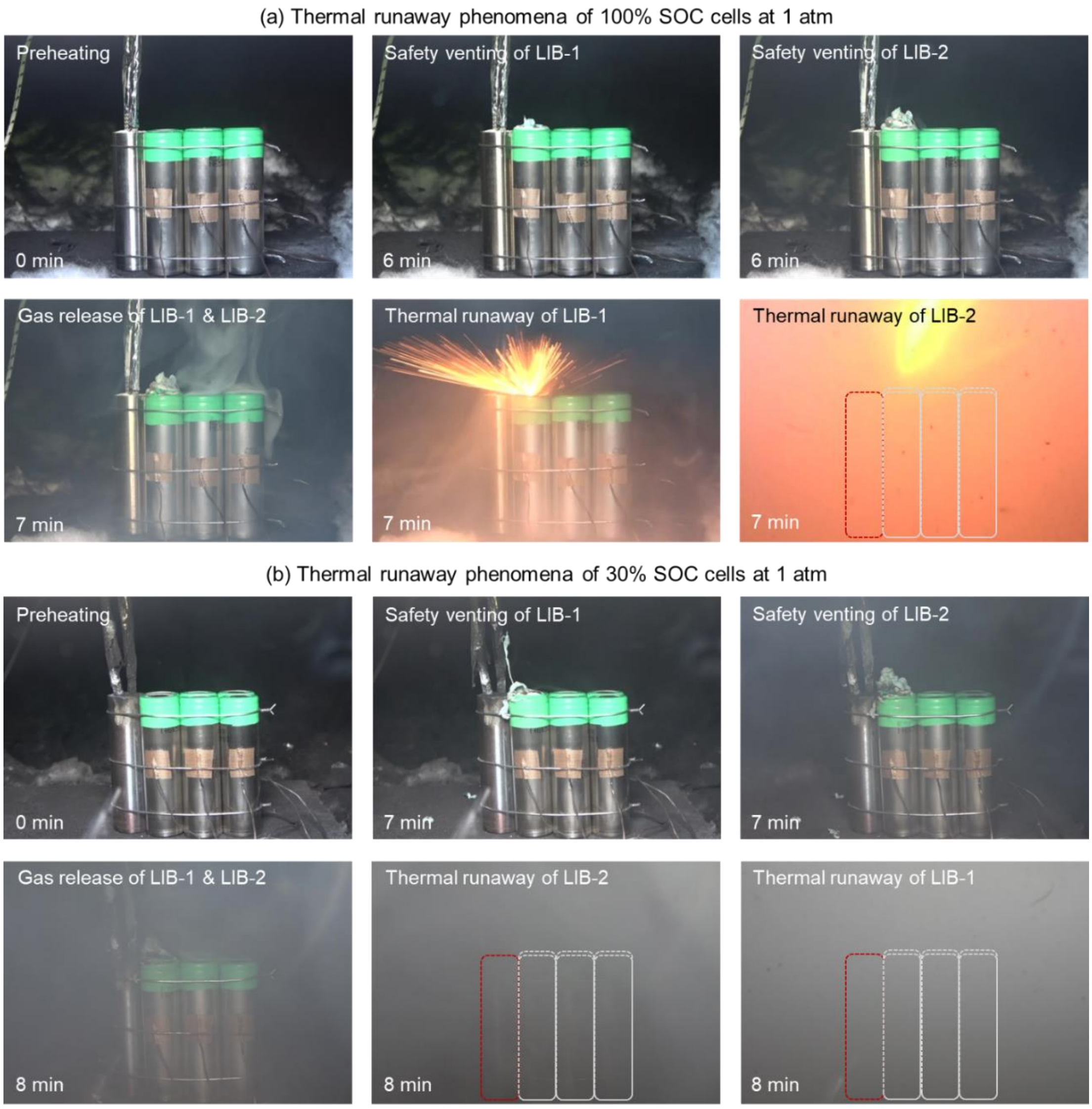

Fig. 2. Thermal runaway phenomena of the Layer-I cells with the SOC level of (a) 100% (Video S1) and (b) 30% (Video S2) at 1 atm.

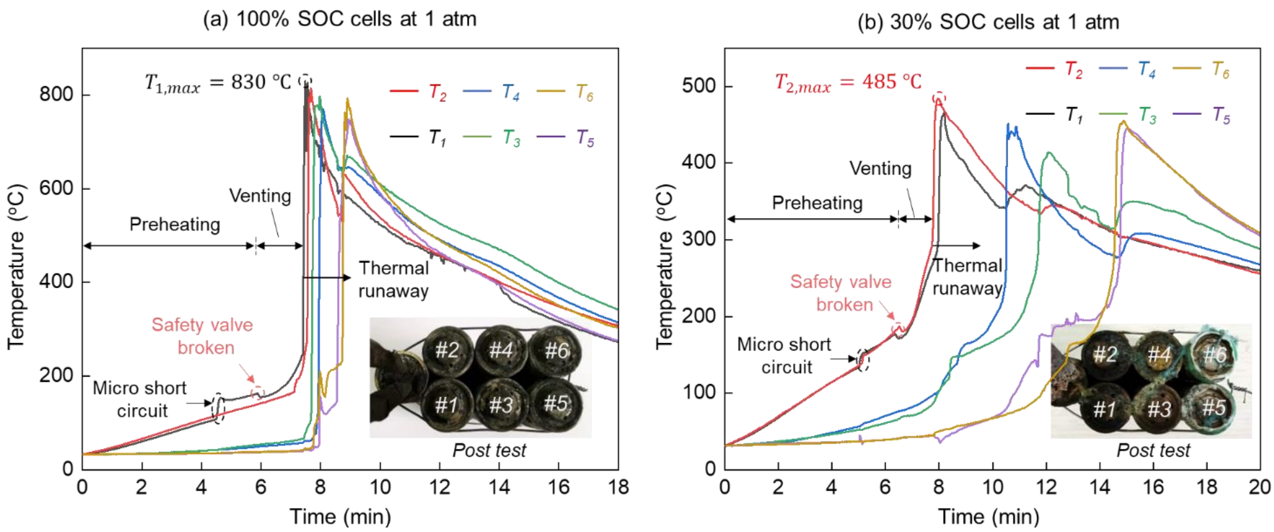

Fig. 3. Temperature evolution curves of battery piles with (a) 100% SOC and (b) 30% SOC at 1 atm.

To understand the thermal runaway and its propagation behavior, Fig. 3 presents the temperature-evolution curves of LIB piles at 1 atm (the same tests in Fig. 2). As shown in Fig. 3a, LIB-1 is the first cell that goes to thermal runaway. According to the temperature evolution, its thermal runaway process can be divided into three stages: (I) preheating, (II) venting, and (III) thermal runaway, agreeing with the previous thermal-runaway studies [46]. Under the external heating, the temperature of LIB-1 gradually increases with an average rate of 0.2 °C/s. After 4min, its temperature rapidly increases from 123.7 °C to 153.2 °C due to the occurrence of the internal micro short circuit (MSC). The MSC is a common phenomenon for batteries in heating abuse conditions, which is ascribed as the separator flaws. At around 6 min, a slight temperature fluctuation appeals, indicating the occurrence of safety venting. The temporary cooling caused by safety venting cannot stop the internal reactions. Conversely, the battery temperature continues to rise with an average rate of 0.4 °C/s during the venting stage. With the battery temperature increasing, the internal chemical reaction rate could be accelerated. As a consequence, the thermal runaway of LIB-1 occurs where its temperature sharply rises to 830 °C at a maximum rate of about 70 °C/s. After around 8 s, LIB-2 reaches thermal runaway. Since thermal runaway of Layer-I cells can release large amounts of heat, the thermal runaway of Layer-II cells (LIB-3 and LIB-4) was quickly triggered (Fig. 3a) within 20 s. And the propagation time for thermal runaway from Layer II to Layer III was about 40 s.

In terms of 30% SOC cells, the thermal runaway process also contains three stages of preheating, safety venting, and thermal runaway. As depicted in Fig. 3b, LIB-2 is the first cell that reaches thermal runaway. Although there is no flaming combustion (Fig. 2b), the maximum battery surface temperature of LIB-2 can still exceed the 485 °C. According to the temperature curves for the entire battery pile, a 1-D layer-to-layer propagation can be observed. Specifically, the average propagation time for thermal runaway from Layer I to Layer II was about 190 s, while the propagation time from Layer II to Layer III was about 218 s. As expected, the overall thermal runaway propagation time of 30% SOC battery pile is 6.5 times longer than that of 100% SOC battery pile, showing a lower fire risk.

3.2 Thermal runaway propagation in low pressure

To further investigate the characteristics of thermal runaway propagation, Fig. 4 shows the temperature evolution curves of LIB piles in low pressure environment. For 100% SOC cells at 0.6 atm (Fig. 4a), a clear 1-D layer-to-layer propagation is shown. The thermal runaway of LIB-1 and LIB-2 occurred almost at the same time. The thermal runaway of Layer-I cells occurred at about 456 s, causing a sudden pressure increase of 13 kPa. The thermal runaway propagation time from Layer I to Layer II (Δt_1) is 67 s, and the time interval of thermal runaway between Layer II and Layer III (Δt_2) is approximately 85 s. The explosion happened after the thermal runaway of Layer-III cells. Such an explosion was caused by LIB-6 rupture (ejection of jelly rolling) and only occurred once in this work. This special phenomenon has been reported in the previous studies and is prone to happen in the enclosed environment [47] or inert environment [43]. The overall time for thermal runaway propagation at 0.6 atm is prolonged by 90 s compared with the propagation time at 1 atm. For 30% SOC cells at 0.6 atm (Fig. 4b), the layer-to-layer propagation is also obvious. The thermal runaway of two cells in a layer can lead to a pressure increase of 1.5 kPa, much lower than that of 100% SOC cells. In this case, the total propagation time for thermal runaway is 460 s, 52 s longer than that of 30% SOC cells at 1 atm. Fig. 4c shows the evolution curves of temperature and ambient pressure for 100% SOC cells at 0 atm. It is worthy to note that no flaming combustion was observed during the test (Video S3). As the oxygen content at 0 atm is near 0, this phenomenon suggests that the concentration of ejected gases cannot meet the lower flammability limit, agreeing with the results reported by previous studies [14,34]. Nevertheless, these ejected gases can lead to an increase in ambient pressure of about 2 kPa. After thermal runaway at 0 atm, the maximum temperature for Layer-I cells is 736 °C, about 100 °C lower than the maximum temperature at 1 atm (Fig. 3a). Compared to the thermal runaway propagation time for 100% SOC cells at 1 atm (Fig. 3a), the overall time is prolonged by 150% at 0.6 atm and 172% at 0 atm. Thus, ambient pressure shows a big effect on thermal runaway propagation.

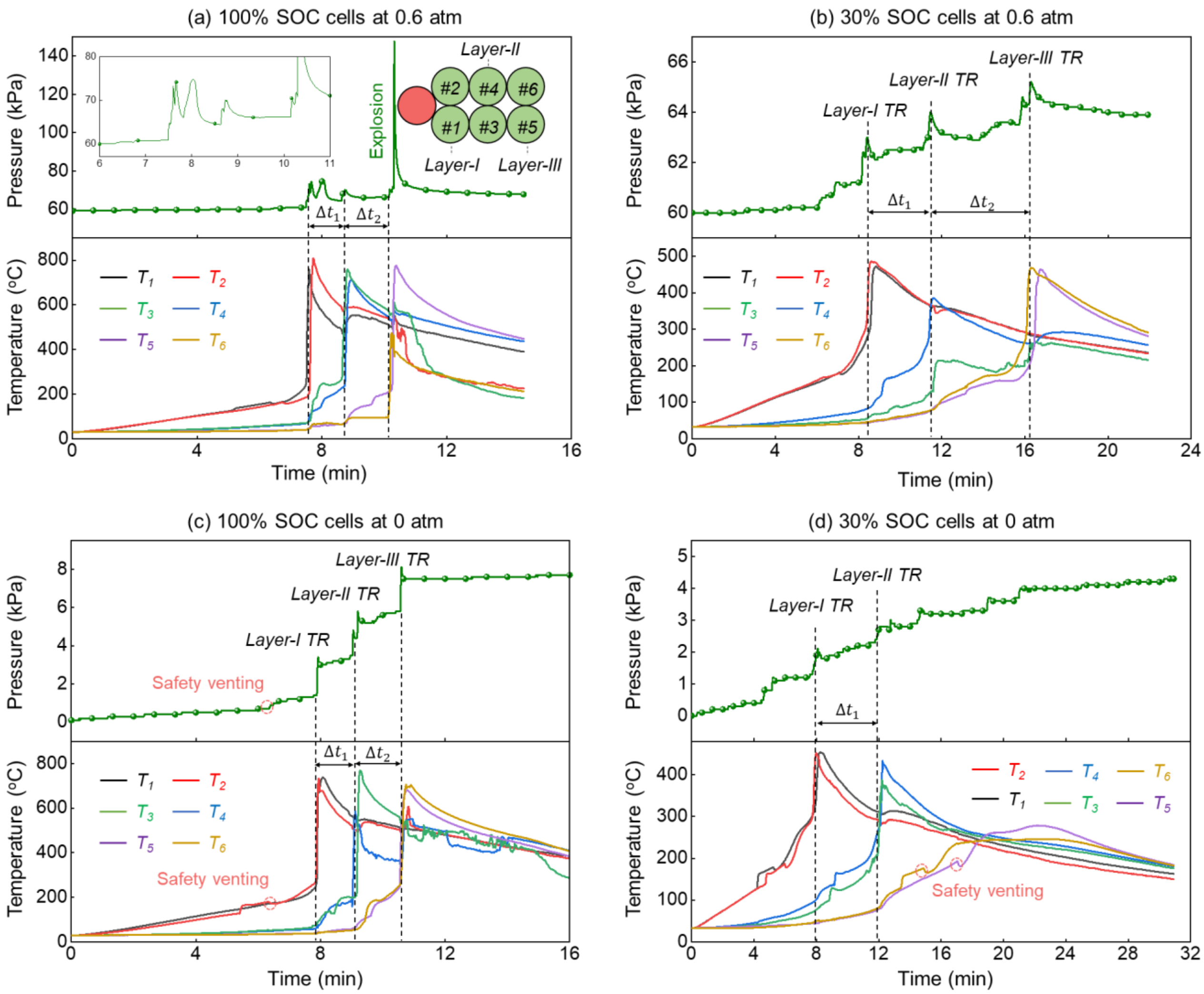

Fig. 4. Evolution curves of battery temperature and ambient pressure for cells with (a) 100% SOC at 0.6 atm, (b) 30% SOC at 0.6 atm, (c)100% SOC at 0 atm, and (d) 30% SOC at 0 atm.

For 30% SOC cells at 0 atm, it is interesting to see that thermal runaway could not propagate through the entire battery pile (Fig. 4d). Specifically, the thermal runaway of the Layer-I cells is triggered at around 8 min with a maximum surface temperature of 480 °C. Then, it costs about 256 s for the thermal runaway to propagate from Layer I to Layer II (Δt_1), exhibiting a slower propagation speed compared with 30% SOC cells at 1 atm and 0.6 atm. For Layer-III cells, although the safety venting happened, the thermal runway was not triggered. This is because the heat accumulation inside the Layer is insufficient to raise its temperature to the thermal runaway threshold. Due to the weak thermal runaway reactions, the thermal runaway for a 30% SOC cell could only cause the ambient pressure to increase by less than 1 kPa. In short, both the physical observations and temperature information indicate that cells reducing the ambient pressure can lower the battery thermal hazards and slow down the thermal runaway propagation speed. In conventional cognition, the delay of thermal runaway propagation could be attributed to the weakening of the flaming combustion. However, no jet flame was formed during the thermal runaway of 30% SOC cells in all cases. Thus, there would be another reason brought by low ambient pressure that alleviates the thermal runaway propagation in the 30% SOC LIB piles.

3.3 Pressure effect on LIB temperatures and mass loss

In this work, the battery surface temperature is used to capture the thermal runaway propagation process and reflect the corresponding characteristics. Herein, the safety venting temperature (T_v), thermal runaway onset temperature (T_{TR}), and maximum surface temperature after thermal runaway (T_{max}) are employed to analyze the pressure effect on thermal runaway propagation. Safety venting (or safety valve bursting) is an important signal for battery thermal failure. T_v was determined as the earliest point where the temperature derivative (dT/dt) became negative. Thermal runaway onset temperature (T_{TR}) is associated with the battery thermal stability. In this work, T_{TR} was identified as the temperature point where the value of dT/dt exceeded the 10 °C/s. This criterion considered the gradient of battery surface temperature caused by side heating and was established by combing visual and audible observations, which was adopted in many studies [12]. As for the maximum surface temperature (T_{max}), it is related to the intensity of exothermal reactions inside the cell and indicates the thermal hazards of battery thermal runaway. During the experiments, several thermocouples fell off from the cell surface, so these data have been excluded from the discussion.

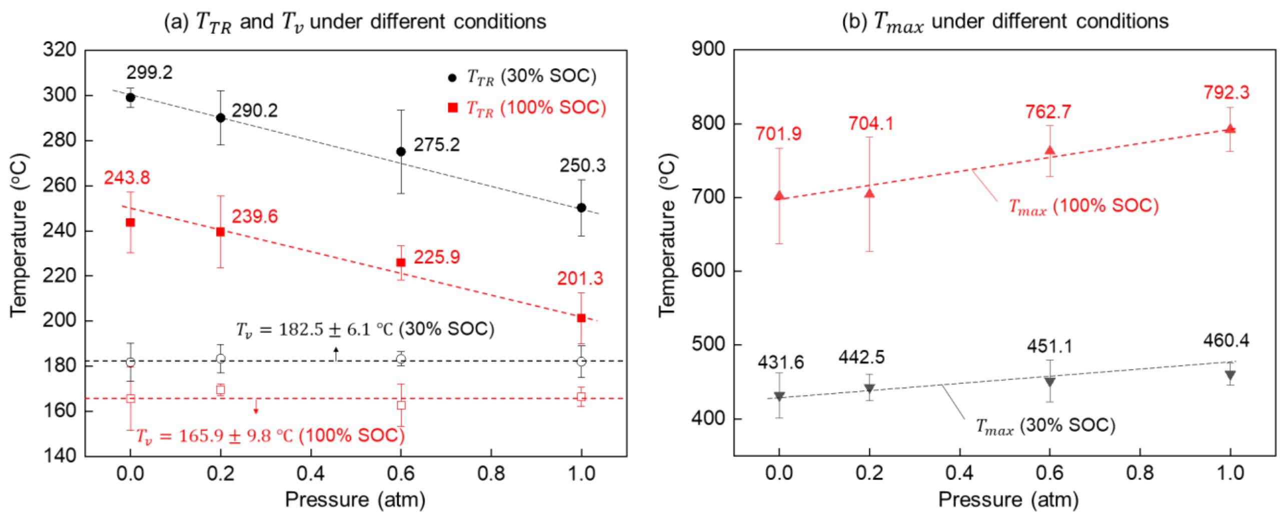

Fig. 5. The characteristic temperatures of (a) safety venting (T_v) and thermal runaway (T_{TR}), as well as the maximum surface temperature (T_{max}) under different conditions.

Fig. 5 summarizes the characteristic temperatures of T_v, T_{TR}, and T_{max} under different conditions. It is evident that the safety venting temperature (T_v) is insensitive to the ambient pressure but increases with the SOC level (Fig. 5a). This is because the safety venting happens when the internal pressure reaches a certain threshold, around 15~20 atm (1.5~2 MPa) [43,44]. This threshold is much higher than the pressure gradient (0.02~0.04 MPa) selected in this work. Therefore, the safety venting strongly depends on internal reactions and is insensitive to the ambient pressure. Specifically, T_v is 182.5 ± 6.1 °C and 165.9 ± 9.8 °C for 30% SOC and 100% SOC cells, respectively. As for thermal runaway temperature (T_{TR}), it decreases as the SOC level increases or ambient pressure decreases. For example, T_{TR} for 100% SOC cells is about 50 °C lower than that of 30% SOC cells, showing a higher fire risk. The value of T_{TR} for 100% SOC cells decreases from 243.8 °C to 201.3 °C as the ambient pressure increases from 0 atm to 1 atm. In other words, it is more difficult to initiate the thermal runaway under low ambient pressure. The gas flow was choked when the safety valve opened and would return to subsonic levels until the onset of thermal runaway. During the venting stage, the internal pressure of the cell is approximately equal to the ambient pressure until the beginning of the thermal runaway [44]. Thus, the evaporation of electrolytes, solvents, and binders will be accelerated under low ambient pressure, reducing reactants involved in the exothermic reactions. That is why the cells are difficult to accumulate heat to trigger thermal runaway at low pressure, resulting in the higher thermal runaway temperature (T_{TR}). Consequently, a lower maximum surface temperature (T_{max}) is performed for cells under low pressure (Fig. 5b). Besides, the T_{max} of 100% SOC cells is much higher than that of 30% SOC cells, indicating higher fire hazards. In short, the ambient pressure has a significant impact on the value of T_{TR}, and T_{max}, while the pressure effect on T_v is negligible. Due to the weak exothermic reactions inside the cell, a higher T_{TR} and a lower T_{max} exhibit under low ambient pressure.

Before and after the experiments, the mass of each cell was measured and recorded. To intuitively compare the mass change, the mass loss fraction is calculated and illustrated in Fig. 6a. As expected, the mass loss fraction increases as the SOC increases, consistent with the observation that more violent thermal runaway occurred at a higher SOC. Specifically, the mass loss fraction increases from 14.6% (~6.0 g) to 31.5% (~12.9 g) as the SOC level increases from 30% to 100% at 1 atm. Moreover, the mass loss fraction decreases slightly with ambient pressure, agreeing with the experimental results presented by previous works [41]. As the ambient pressure decreases from 1 atm to 0 atm, the mass loss fraction for 100% SOC cells decreases from 31.5% (~12.9 g) to 26.8% (~11.0 g). Therefore, the mass loss results further verified the mild thermal runaway reaction for cells under low ambient pressure.

The mass loss for the 18650-type cell after thermal runaway depends on the cathode chemistry, cell capacity, and heating method. For example, the Ni-rich cells always have a violent thermal runaway and a higher mass loss [48]. Similarly, cells with higher capacity and heating power also exhibited higher mass loss [41,49]. Additionally, the external heating tests could experience a higher mass loss than internal heating tests [46]. As for external heating, the mass loss of cells with the surrounding heating coil is higher than that of cells heated by the side heater [50]. That is why the mass loss of a 2.5 Ah NMC532 cell (100% SOC) under 125-W coil heating (25 g) [43] is higher than that of the 2.01 Ah NMC532 cell under 300-W side heating (18 g) [41], the 2.2 Ah NMC 532 cell under 100-W side heating (13 g) in this work, and the 2.2 Ah NMC cell under 20-W wire heating (11 g) [49]. It should be noted that the stripping of the cell plastic packaging (~2 g) might also contribute to the relatively lower mass loss of the last two values, as the plastic packaging would burn out during the thermal runaway process.

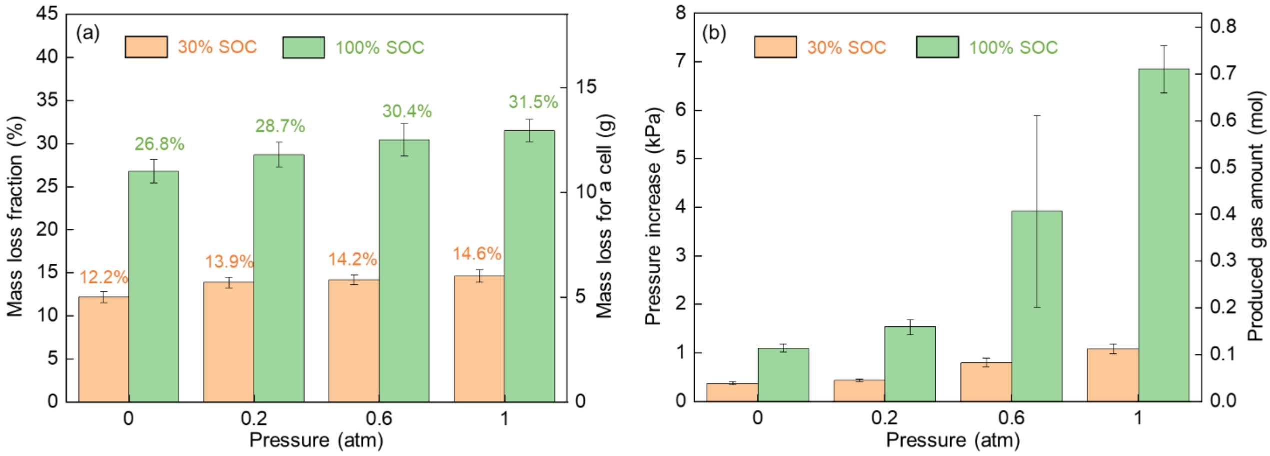

Fig. 6. (a) Mass loss fraction of cells after the thermal runaway and (b) ambient-pressure increase caused by thermal runaway of a cell.

When thermal runaway occurred, the ejected gases and aerosol would cause the pressure increasing inside the chamber. The value of ambient pressure increase (ΔP) is an indicator to reflect the intensity of thermal runaway reactions, which can be used to estimate the amount of gas products during the thermal runaway. Assuming that the emitted gases satisfy the ideal gas law, the amount of produced gases could be estimated as

$$n_{gas} = \frac{\Delta PV}{RT_{gas}} \tag{1}$$

Where n_{gas} is the amount of gases generated from the thermal runaway of single cell, ΔP is the calculated pressure-increase of chamber induced by thermal runaway, $V = 0.26\, m^3$ stands for the volume of low-pressure chamber, $R = 8.314\ \mathrm{J/K - mol}$ is the ideal gas constant, and T_{gas} is the temperature of gases that can be regarded as the chamber temperature T_a. As the gases temperature was not measured in the tests, the gases temperature can be regarded as the T_{max} [51]. Thus, the ambient pressure increase (ΔP) and calculated gas amount (n_{gas}) are presented in Fig. 6b. Although the gas amount at 0.6 atm exhibits a relatively large uncertainty due to high variability of cells during the production process, the values of ΔP and n_{gas} increase with the SOC and ambient pressure. For a 100% SOC cell, the released gas is 0.71 mol gas during the thermal runaway at the standard pressure (1 atm), while it decreases to 0.11 mol at 0 atm. Therefore, the intensity of thermal runaway is weak under the low ambient pressure, which will be further verified by X-ray imaging and discussed in the following section.

3.4 X-ray CT imaging for LIB debris

Computed tomography (CT) scanning and X-ray radiography are non-destructive technologies that can be used to detect the internal structure changes of the aged LIBs [38] or burnt LIBs [35]. There are many X-ray CT images for fresh 18650-type cells in the literature, where the winding structures of the electrodes are well organized [52]. In this work, X-ray CT images of the burnt cells (after thermal runaway) at 0.2 atm and 1 atm are exhibited in Fig. 7. After thermal runaway, the winding structures of the electrodes are disorganized, and many particles can be observed within the electrode plates. The change in electrode structure is attributed to the gas generation and ejection inside the cell. And the fine particles could be molten drops of the metal aluminum that was used for the current collector of cathode materials. The melting temperature for aluminum is around 660 °C, which is much lower than the maximum surface temperature of 100% SOC cells. It should be noted that the disproportionation and decomposition of the cathode materials contribute to the major heat during thermal runaway, and the ejected gases are mainly released from the cathode reactions. Thus, the degree of structural looseness and the number of metal particles can be employed to indicate the intensity of the internal thermal runaway reactions.

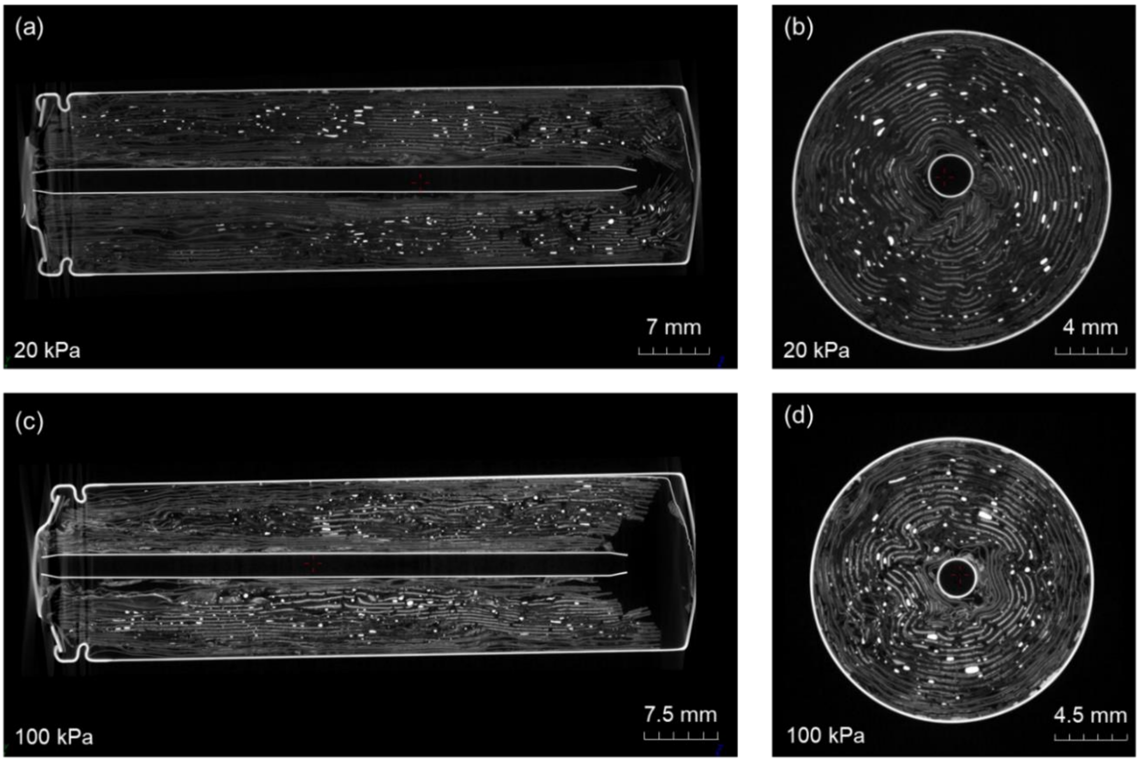

Fig. 7. CT scan of 100% SOC cells after thermal runaway: (a) lengthwise cross-section and (b) axial cross-section at 0.2 atm; (c) lengthwise cross-section and (d) axial cross-section at 1 atm.

Compared to the images of the lengthwise cross-section between 0.2 atm (Fig. 7a) and 1 atm (Fig. 7c), a looser structure of electrodes can be found in the cell at 1 atm. For axial cross-section images (Fig. 7b and Fig. 7d), more mental particles can be observed in the cell at 1 atm. Apparently, cells at higher ambient pressure have a stronger thermal runaway reaction. During the venting stage, the internal pressure of the cell is approximately equal to the ambient pressure until the beginning of the thermal runaway [44]. At this stage, the evaporation of electrolytes, solvents, and binders will be accelerated under low ambient pressure, reducing reactants involved in exothermic reactions. Thus, exothermic reactions in the venting stage are weak under low ambient pressure.

These results are consistent with the surface-morphology results for cathode materials and the ejected particles in previous studies [53]. Specifically, NASA analyzed the thermal runaway particles using the energy dispersive X-ray (SEM/EDX), Fourier-transform infrared spectroscopy (FTIR), and Gas chromatography–mass spectrometry (GC-MS). They found that the volatile electrolyte components had a higher percentage under vacuum conditions [53]. Therefore, the results of X-ray CT imaging further confirm that the overall exothermic reactions are weaker under low ambient pressure, resulting in higher thermal runaway temperature (T_{TR}), lower maximum surface temperature (T_{max}), and lower gas production (Fig. 5 and Fig. 6).

3.5 *Analysis of thermal runaway propagation rate*

Thermal runaway propagation rate (or speed) is extensively used in studies to quantify the development of thermal hazards in a LIB module [32]. Based on the structure of the LIB pile in this work, a 1-D (linear) layer-to-layer propagation rate for the thermal runaway is defined, which represents the number of layers that undergo thermal runaway per minute [layer/min] as

$$r = \frac{n}{\Delta t} \tag{2}$$

where Δt [min] is the total time of thermal-runaway propagation, and n [layer] is the number of layers that undergo thermal runaway. It just aims to describe the propagation level of thermal runaway and quantify the development of thermal hazards in the battery pile. According to the experimental data, the calculated thermal runaway propagation rates (r) varying with the ambient pressure (P) are summarized in Fig. 8a.

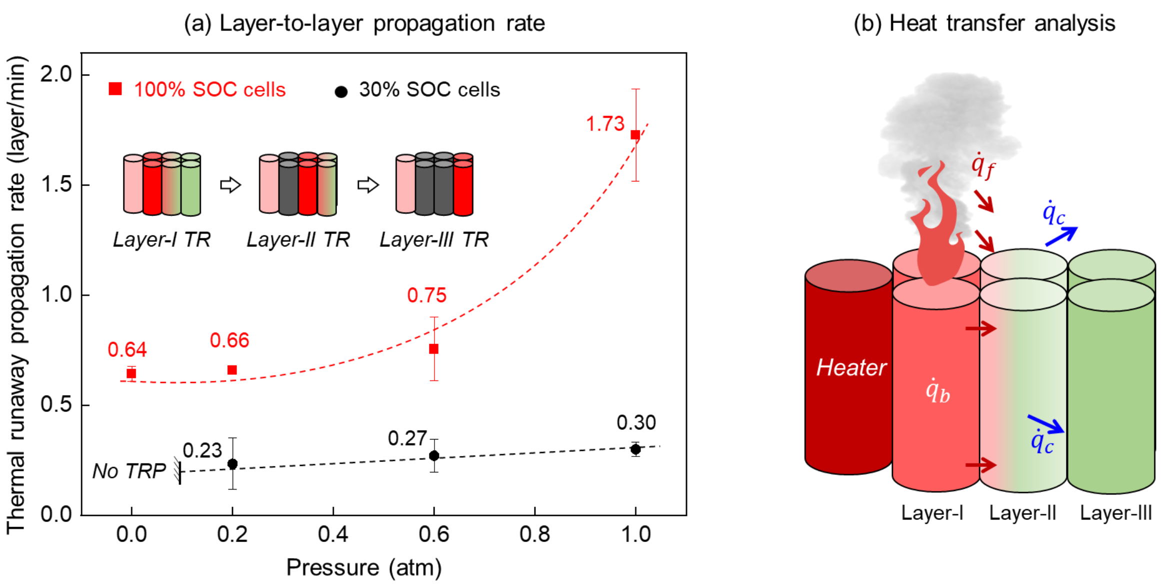

Fig. 8. (a) Thermal runaway propagation rate in LIB piles under different pressures, and (b) schematic of the thermal runaway propagation in the LIB pile.

Apparently, the thermal-runaway propagation rate increases with both SOC level and pressure value. At 1 atm, thermal runaway propagation rate for 100% SOC cells is 1.73 [layer/min], 5 times higher than of 30% SOC cells. As the ambient pressure decreases from 1 atm to 0 atm, the thermal runaway propagation rate for 100%

SOC cells decreases from 1.73 to 0.64 [layer/min], with a decrement of 63%. And the thermal runaway cannot propagate in the 30% SOC LIB piles at 0 atm. In literature, thermal runaway propagation of 100% SOC NCM has been tested in the both air and nitrogen environment [12]. Filling battery enclosures with nitrogen gas, the row-to-row propagation rate for thermal runaway can be reduced by 39%, lower than decrement of 63% in this work. Therefore, reducing the ambient pressure can significantly mitigate the thermal runaway propagation in open-circuit battery piles, increasing the overall safety for LIBs in storage and transport.

Similar to the fire spread process [54], the thermal-runaway propagation can be regarded as a continuous ignition process of cells. The sufficient condition for thermal runaway to propagate from layer to layer is that heat released by failed layers can increases the temperature of next-layer cells to the thermal runaway temperature (T_{TR}). For cells in the failed layer, the thermal hazards consist of the internal heat generation and external flaming combustion or hot-material ejection (Fig. 8b). The heat transfer between two neighboring layers includes two modes: heating from external flame or smoke ($\dot{q}_f$) and heating transfer within the battery pile ($\dot{q}_b$). Although the jetting flame can affect the thermal runaway propagation, the energy flow calculation in previous studies suggested that thermal runaway propagation is dominated by heat conduction between cells [41,55]. To qualitatively analyze the effect of ambient pressure on thermal runaway propagation, a simplified heat transfer model is performed. Assuming that battery layer is internally uniform, the overall propagation rate (r) is controlled by the ratio of net heating rate to the thermal inertia of the cell as

$$r = \frac{\text{Net heating rate}}{\text{Energy required for initiating thermal runaway}} \approx \frac{\dot{q}_b - \dot{q}_c}{m_{LIB} c_{LIB} (T_{TR} - T_a)} \tag{3}$$

where $\dot{q}_b$ is the effective heating rate through cells, $\dot{q}_c$ is the environmental cooling rate, m_{LIB} is the mass of the LIB, and c_{LIB} is the total heat capacity of the LIB. Considering the near-limit condition for thermal runaway propagation ($T_{layer-1} \to T_{max}$), Eq. (3) can be rewritten as

$$r \approx \frac{h_{TR} A_{LIB} (T_{max} - T_{TR}) - h_c A_{LIB} (T_{TR} - T_a)}{m_{LIB} c_{LIB} (T_{TR} - T_a)} = \frac{h_{TR} A_{LIB} \frac{T_{max} - T_{TR}}{T_{TR} - T_a} - h_c A_{LIB}}{m_{LIB} c_{LIB}} \tag{4}$$

where h_{TR} is the effective heat transfer coefficient between layers, A_{LIB} is the cell surface area, and h_c is the effective cooling coefficient which can be lowered by flame. Defining a dimensionless number

$$T^* = \frac{T_{max} - T_{TR}}{T_{TR} - T_a} = \frac{\text{Overall heating}}{\text{LIB thermal inertia}} \tag{5}$$

Fig. 9 summarizes the value of T^* based on the experimental data in Fig. 5. Clearly, T^* increases with both SOC level and ambient pressure value. The good linear fitting indicates that the value of T^* is proportionate to the ambient pressure as $T^* \propto P$.

The convective cooling coefficient (h_c) in Eq. (3) can be expressed as

$$h_c = Nu \frac{k_a}{L} \propto Gr^m \propto \rho^{2m} \propto P^{2m} \quad (m > 0) \tag{6}$$

where Nu is the Nusselt number; Gr is the Grashof number; k_a is the conductivity of air; and L is characteristic length, respectively; ρ is the air density. Thus, as the pressure decreases, the convective coefficient (h_c) and environmental cooling decreases. According to our previous study [39], the relationship between effective cooling coefficient (h_c) and ambient pressure should be $h_c \propto P^{0.2}$.

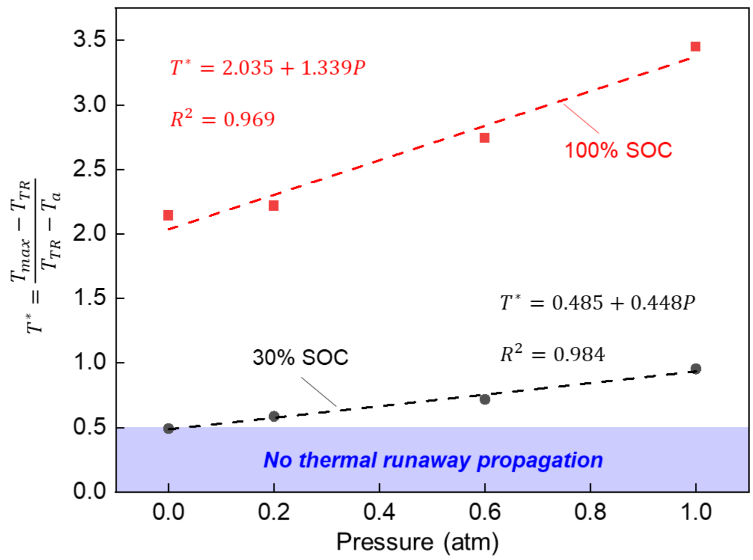

Fig. 9. The dimensionless temperature T^* varies with the ambient pressure P.

Compared with the pressure effect on the overall intensity of thermal runaway reactions ($T^* \propto P$), the pressure effect on the effective cooling coefficient (h_c) is negligible. Therefore, Eq. (4) can help to qualitatively describe the effects of SOC and ambient pressure (P) on thermal runaway propagation rate. Specifically, the thermal runaway propagation rate (r) increases with (T^*), which increases with the SOC or ambient pressure (P). In other words, thermal runaway propagation will be postponed by reducing SOC level or ambient pressure.

Moreover, Eq. (4) defines the limit conditions for thermal runaway propagation in such a layer-based battery pile. To prevent the thermal runaway propagation ($r \leq 0$), the dimensionless T^* should satisfy

$$T^* = \frac{T_{max} - T_{TR}}{T_{TR} - T_a} \leq \frac{h_c}{h_{TR}} \tag{7}$$

Firstly, there should be a maximum ambient temperature (T_a) for open-circuit cells to prevent the thermal runaway propagation, which is given by

$$T_{a,max} = T_{TR} - \frac{h_{TR}}{h_c}(T_{max} - T_{TR}) \tag{8}$$

Therefore, Eq. (8) further verifies the importance of ambient temperature for LIBs in safe storage [56] and can help to evaluate the safety of the storage conditions from the perspective of mitigating thermal runaway propagation. In the real scenario, thermal insulation materials are inserted between cells, leading to a lower h_{TR}. To suppress the thermal runaway propagation, there is a maximum value for the effective heating coefficient (h_{TR}), which can be expressed as

$$h_{TR,max} = \frac{T_{TR} - T_a}{T_{max} - T_{TR}} h_c \tag{9a}$$

Meanwhile, a higher h_c can lead to a lower thermal runaway propagation rate. Similarly, the minimum value for the effective cooling coefficient from the environment (h_c) can be expressed as

$$h_{c,min} = \frac{T_{max} - T_{TR}}{T_{TR} - T_a} h_{TR} \tag{9b}$$

As discussed above, reducing the ambient pressure can alleviate the thermal runaway propagation due to the weakening of both external flaming combustion and internal chemical reactions, which is qualitatively described by a simplified heat transfer analysis. Additionally, the proposed heat transfer model can help to determine the key points for safe storage and transport of LIBs, such as ambient temperature, effective thermal insulation strategies, and effective cooling methods.

3.6 Comparison with literature data

In literature, the layer-to-layer propagation for thermal runaway of NCM cells has been investigated in several works [12,23,35,41,57,58]. A comparison is delivered based on the literature data to highlight the pressure effect and figure out other factors on thermal runaway propagation, as shown in Fig. 10. For example, Weng et al. [35] studied the thermal runaway propagation of 75%-SOC NCM cells in a triangle module at 0. 95 atm. Compared to thermal runaway propagation in the current work, the 2-D layer-to-layer propagation in their study is slower. This is because both the capacity (2.0 Ah) and SOC (75%) of their LIB sample are lower. Meanwhile, the layer-to-layer propagation rate also depends on the cathode material of LIB. Liu et al. [57] conducted thermal runaway propagation tests in the NCM111 LIB pile. Due to the lower nickel content inside the battery, thermal runaway intensity is weak, resulting in a lower thermal runaway propagation rate. Moreover, the configuration of the battery module has a significant influence on thermal runaway propagation. Zhong et al. [58] studied the thermal runaway propagation in a 3×3 LIB pile with a 200-W cylindrical heater. The 1-D layer-to-layer propagation is much slower than that in 2×3 LIB pile due to the more cells inside the LIB pile.

Recently, Jia et al. [41] conducted the thermal runaway propagation test under the ambient pressure of 0.95, 0.7, and 0.35 atm. The LIB sample used in their work was the 18650-type cell with $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$ (NCM523) as cathode material. Thermal runaway propagation was initiated by a 300-W heating rod with the same dimension as LIB sample. Because of the assymetrical array, the studied layer-to-layer propagation was not perfectly 1-D but more close to a 2-D process. For comparison, their 2-D layer-to-layer propagation rate is calculated and depicted in Fig. 10. As the ambient pressure decreases, the thermal runaway propagation decreases, agreeing with the experimental results in this work. However, the propagation rate in their work is much higher, which could be attributed to the higher initial heating power and the different module configurations. Thus, more studies are needed to explore the effects of LIB chemistry, heating power, and module configuration effects on thermal runaway propagation under low ambient pressure.

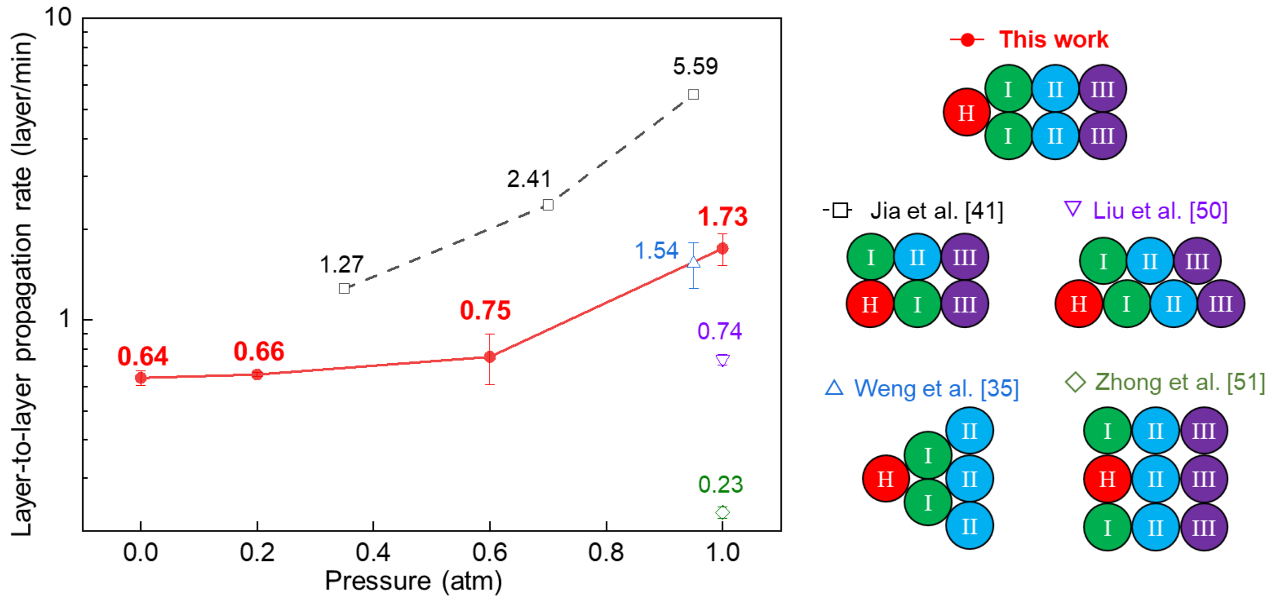

Fig. 10. A comparison of layer-to-layer propagation rate for thermal runaway in 100% SOC NCM batteries, where 'H' indicates the heater, and 'I, II, and III' indicate the layer number.

Although the layer-to-layer propagation has been compared with literature data, this work still has several limitations. First, cells were highly contacted with each other in the current study to simulate the worst scenario. In actual storage and transport, thin cardboard or plastic barriers are inserted between cells and would change the thermal runaway propagation behaviors. Furthermore, only horizontal thermal runaway propagation was considered in this work, whereas the flame jetting could trigger the vertical thermal runaway propagation in the battery stockpiles [59,60]. Moreover, the capacity of the 18650-type cell (ICR18650–22F, 2.2 Ah) used in this work is slightly lower than that of the current popular cell (INR18650−35E, 3.5 Ah). Thus, future works will study the effects of insulation materials and cell capacity on thermal runaway propagation in the horizontal and vertical directions under low ambient pressure.

4. Conclusions

This work investigates the thermal runaway propagation for 18650-type cells with two SOC levels (30% and 100%) and four ambient pressures ranging from 0 atm to 1 atm. The thermal runaway process of a single cell includes three stages of preheating, venting, and thermal runaway. Remarkably, the venting stage can be regarded as an incubation period for thermal runaway. Reducing the ambient pressure, the evaporation of electrolytes, solvents, and binders will be accelerated at this stage, decreasing the amount of reactants in the exothermic reactions. Thus, the overall thermal runaway reactions are weak under low pressure, leading to the higher thermal runaway temperature (T_{TR}), lower maximum surface temperature (T_{max}), lower mass loss, and lower gas production. Such a pressure effect has been verified by X-ray CT imaging for LIB debris and compared with the literature data.

To quantify the development of thermal hazards in the battery pile, a layer-based thermal runaway propagation rate is proposed and calculated based on experimental results. The thermal runaway propagation rate increases with ambient pressure and SOC level. For 100% SOC cells, the thermal runaway propagation rate increases from 0.64 [layer/min] to 1.73 [layer/min] as the ambient pressure increases from 0 atm to 1 atm, with an increment of 170%. At 1 atm, the increment of thermal runaway propagation rate is about 476% as the SOC increases from 30% to 100%. Moreover, a simplified heat transfer model is proposed to qualitatively describe the effects of SOC and ambient pressure on thermal runaway propagation. Future work will study the effects of insulation materials and cell capacity on thermal runaway propagation in the horizontal and vertical directions, to assess the fire risks in the real storage scenario under low pressure.

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CRediT authorship contribution statement

Yanhui Liu: Data curation, Investigation, Writing - Original Draft, Formal analysis.

Huichang Niu: Resources, Supervision.

Jing Liu: Investigation, Resources, Funding acquisition.

Xinyan Huang: Conceptualization, Methodology, Funding acquisition, Supervision, Writing - Review & Editing.

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